

PROGRAM 4: BREADTH FIRST SEARCH (BFS)

Aim

To write a Python program to implement Breadth First Search (BFS) traversal of a graph.

Objective

To traverse all the vertices of a graph level by level using the BFS algorithm.

Algorithm

1. Start.
2. Create a graph using an adjacency list.
3. Mark all vertices as unvisited.
4. Insert the starting vertex into a queue.
5. Remove a vertex from the queue and print it.
6. Add all unvisited adjacent vertices to the queue.
7. Repeat until the queue becomes empty.
8. Stop.

Source Code

```
graph = {  
    'A': ['B', 'C'],  
    'B': ['D', 'E'],  
    'C': ['F'],  
    'D': [],  
    'E': [],  
    'F': []
```

```
}
```

```
visited = []
```

```
queue = []
```

```
visited.append('A')
```

```
queue.append('A')
```

```
while queue:
```

```
    node = queue.pop(0)
```

```
    print(node, end=" ")
```

```
    for i in graph[node]:
```

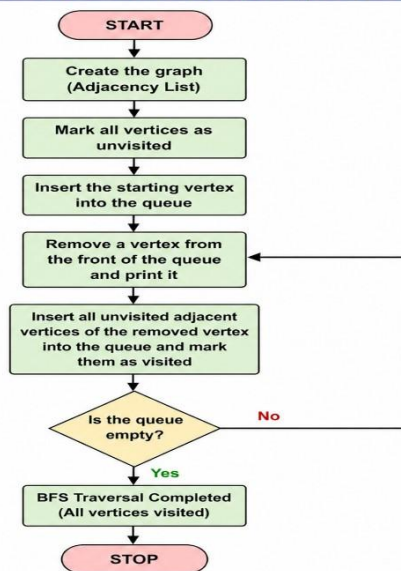
```
        if i not in visited:
```

```
            visited.append(i)
```

```
            queue.append(i)
```

Flowchart

BREADTH FIRST SEARCH (BFS) FLOWCHART



Sample Output

BFS Traversal:

A B C D E F

```
>>> |==== RESTART: C:/User
>>> |A B C D E F
```

Result

Thus, the Python program to implement Breadth First Search (BFS) was executed successfully.